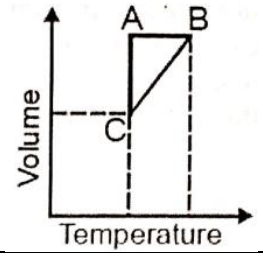
Question	A well-stoppered thermos flask contains some ice cubes. This is an example of a:
Type	multiple choice
Option	Closed system
Option	Open system
Option	Isolated system
Option	Non-thermodynamic system
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is not a state function:
Type	multiple choice
Option	Internal energy
Option	Enthalpy
Option	Work
Option	Entropy
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is true for an adiabatic process:
Type	multiple choice
Option	$\Delta H = 0$
Option	$\Delta W = 0$
Option	$\Delta Q = 0$
Option	$\Delta V = 0$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Among them intensive property is:
Type	multiple choice
Option	Mass
Option	Volume
Option	Surface tension
Option	Enthalpy
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The process, in which no heat enters or leaves the system, is termed as:
Type	multiple choice
Option	Isochoric
Option	Isobaric
Option	Isothermal
Option	Adiabatic
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In thermodynamics which one of the following is not an intensive property:
Type	multiple choice
Option	Pressure

Option	Density
Option	Volume
Option	Temperature
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The temperature of the system decreases in an:
Type	multiple choice
Option	Adiabatic
Option	Isothermal compression
Option	Isothermal expansion
Option	Adiabatic expansion
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Five moles of a gas is put through a series of changes as shown graphically in a cyclic process the $A \rightarrow B$, $B \rightarrow C$ and $C \rightarrow A$ respectively are: 
Type	multiple choice
Option	Isochoric, Isobaric, Isothermal
Option	Isobaric, Isochoric, Isothermal
Option	Isothermal, Isobaric, Isochoric
Option	Isochoric, Isothermal, Isobaric
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Among the following, the state function(s) is/are: (I) Internal energy (II) Irreversible expansion work (III) Reversible expansion work (IV) Molar enthalpy
Type	multiple choice
Option	II and III
Option	I, II, and III
Option	I and IV
Option	I only
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Both q and w are _____ functions.
Type	multiple choice
Option	State
Option	State, Path
Option	Path, State

Option	Path
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A process is called reversible when:
Type	multiple choice
Option	surrounding and system change are the same
Option	there is no boundary between the system and surrounding
Option	surrounding is always in equilibrium with system
Option	system changes into surroundings spontaneously
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Heat exchanged in a chemical reaction at constant pressure is called:
Type	multiple choice
Option	Internal energy
Option	Enthalpy
Option	Kinetic energy
Option	Free energy
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The internal energy is the heat evolved or absorbed at:
Type	multiple choice
Option	constant pressure
Option	constant volume
Option	constant temperature
Option	constant moles
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	When a solid melts, there is:
Type	multiple choice
Option	No increase in enthalpy
Option	Increase in enthalpy
Option	Decrease in enthalpy
Option	Anything can happen
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	One mole of gas occupying 3 litre volume is expanded against a constant external pressure of one atm to a volume of 15 litre. The work in this process is:
Type	multiple choice
Option	$-1.215\times10^{3}J$
Option	$+12.15\times10^{3}J$
Option	$+121.5\times10^{3}J$
Option	$+1.215\times10^{3}J$

Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Both q and w are _____ functions:
Type	multiple choice
Option	State
Option	State, Path
Option	Path, State
Option	Path
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The work done by a system is 8 J when 40 J of heat is supplied to it. The change in internal energy of the system during the process:
Type	multiple choice
Option	32 J
Option	40 J
Option	48 J
Option	-32 J
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A system has internal energy equal to E_1. 450 J of heat is taken out of it and 600 J of work is done on it. The final energy of the system will be:
Type	multiple choice
Option	$(E_1 + 150)$
Option	$(E_1 + 1050)$
Option	$(E_1 - 150)$
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If work done by the system is 300 joule when 100 cal. heat is supplied to it. The change in internal energy during the process is:
Type	multiple choice
Option	-200 joule
Option	400 joule
Option	720 joule
Option	120 joule
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	When 1 mol of a gas is heated at constant volume, temperature is raised from 298 to 308 K. If heat supplied to the gas is 500 J, then which statement is correct?
Type	multiple choice
Option	$q = w = 500 J \ \Delta U = 0$

Option	$q = \Delta U = 500 J \;\; w = 0$
Option	$q = -w = 500 J \;\; \Delta U = 0$
Option	$\Delta U = 0 \; q = w = -500 J$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In a closed insulated container, a liquid is stirred with a paddle to increase its temperature. In this process, which of the following is true:
Type	multiple choice
Option	$\Delta E = W = Q = 0$
Option	$\Delta E \neq 0$, $Q = W = 0$
Option	$\Delta E = W \neq 0$, $Q = 0$
Option	$\Delta E = Q \neq 0$, $W = 0$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A certain electric motor produced 15 kJ of energy each second as mechanical work and lost 2 kJ as heat to the surroundings. What is the change in the internal of the motor:
Type	multiple choice
Option	+17 kJ
Option	+32 kJ
Option	-17 kJ
Option	-28 kJ
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which one of the following equations does not correctly represent the first law of thermodynamics for the given process:
Type	multiple choice
Option	Isothermal process: $q = -w$
Option	Cyclic process: $q = -w$
Option	Isochoric process: $\Delta E = q$
Option	Adiabatic process: $\Delta E = -w$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Change in internal energy, when 4 kJ of work is done on the system and 1 kJ of heat is given out by the system, is:
Type	multiple choice
Option	+1 kJ
Option	-5 kJ
Option	+5 kJ
Option	+3 kJ
Answer	4
Solution	
Positive Marks	4
Negative Marks	1

Question	The internal energy of a substance:
Type	multiple choice
Option	Increases with increase in temperature
Option	Decreases with increase in temperature
Option	Can be calculated by the relation $E = mc^2$
Option	Remains unaffected with change in temperature
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statements about the specific heat of a body is/are correct? 1) It depends upon the mass and shape of the body 2) It is independent of the mass and shape of the body 3) It depends only upon the temperature of the body 129Select the correct answer using the code given below:
Type	multiple choice
Option	1 only
Option	2 and 3
Option	1 and 3
Option	2 only
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	One mole of an ideal monatomic gas requires 210 J heat to raise its temperature by 10 K when heated at constant pressure. If the same gas is heated at constant volume to raise the temperature by the same amount, the heat required is:
Type	multiple choice
Option	108.7 J
Option	115.3 J
Option	129 J
Option	126 J
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If the temperature of 10 g of mercury (of specific heat capacity 140 $Jkg-1k^{-1}$) is to be raised from $0^{\circ}C$ to $100^{\circ}C$ then how much heat (in J) is required?
Type	multiple choice
Option	1400
Option	1.4×10^4
Option	1.4×10^5
Option	140
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If 32 gm of O_2 at $27^{\circ}C$ is mixed with 64 gm of O_2 at $327^{\circ}C$ in an adiabatic vessel, then the final temperature of the mixture will be:
Type	multiple choice
Option	$200^{\circ}C$

Option	$227^{\circ}C$
Option	$314.5^{\circ}C$
Option	$235.5^{\circ}C$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	An electric heater of power 1000 W raises the temperature of 5 kg of liquid from $25^{\circ}C$ to $31^{\circ}C$ in 2 minutes. What is heat capacity of the liquid?
Type	multiple choice
Option	$4\times10^{3}J/kg^{\circ}C$
Option	$2\times10^{4}J/^{\circ}C$
Option	$1.2\times10^{5}J$
Option	$1\times10^{4}J/^{\circ}C$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	For a gaseous reaction, $A(g)+3B(g)\rightarrow 3C(g)+3D(g)$ ΔE is 17 kCal at $27^{\circ}C$ assuming $R=2Cal\ \ K^{-1}$ mol^{-1}, the value of ΔH for the above reaction is:
Type	multiple choice
Option	15.8 Kcal
Option	18.2 Kcal
Option	20.0 Kcal
Option	16.4 Kcal
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	For the reversible isothermal expansion of one mole of an ideal gas at 300 K, from a volume of 10 dm³ to 20 dm^{3}, ΔH is-
Type	multiple choice
Option	1.73 kJ
Option	-1.73 kJ
Option	3.46 kJ
Option	Zero
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The Gibbs free energy change of a reaction at $27^{\circ}C$ is -26 kCal and its entropy change is -60 Cal K. ΔH for the reaction is:
Type	multiple choice
Option	-44 kCals
Option	-18 kCals
Option	34 kals
Option	-24 kCals
Answer	1
Solution	
Positive Marks	4

Negative Marks	1
Question	If S° for H_2, Cl_2 and HCl are 0.13, 0.22 and 0.19 kJ K^{-1} respectively. The total change in standard entropy for the reaction $H_2 + Cl_2 \rightarrow 2HCl\,i$ is:
Type	multiple choice
Option	$30\,JK^{-1}mol^{-1}$
Option	$40JK^{-1}mol^{-1}$
Option	60 $JK^{-1}mol^{-1}$
Option	20 $JK^{-1}mol^{-1}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	When two gases are mixed the entropy:
Type	multiple choice
Option	Remains constant
Option	Decreases
Option	Increases
Option	Becomes zero
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	5 mole of an ideal gas expand reversibly from a volume of 8 dm³ to 80 dm^3 at a temperature of $27^{\circ}C$. The change in entropy is:
Type	multiple choice
Option	41.57 JK-1
Option	$-95.73\,JK^{-1}$
Option	95.73 JK^{-1}
Option	$-41.57\,JK^{-1}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If $\Delta H > 0$ and $\Delta S > 0$, the reaction proceeds spontaneously when:
Type	multiple choice
Option	$\Delta H > 0$
Option	$\Delta H < T\Delta S$
Option	$\Delta H = T\Delta S$
Option	None
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The temperature at which the reaction $Ag_2O(s) \rightarrow 2Ag(s) + \frac{1}{2}O_2(g)$ is at equilibrium is- (Given $\Delta H = 30.5\,kJ\,mol^{-1}$ and $\Delta S = 0.066$ kJ K^{-1} mol^{-1})
Type	multiple choice
Option	462.12 K
Option	362.12 K
Option	262.12 K

Option	562.12 K
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	For the reaction A(s) $\rightarrow \cdot B(s) + C(g)$ the value of ΔH = 30.56 kJ mol^{-1} and $\Delta S = 66\ JK^{-1}$ mol^{-1}. The temperature at which the free energy change for the reaction will be zero is:
Type	multiple choice
Option	373 K
Option	413 K
Option	463 K
Option	493 K
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If the equilibrium constant for a reaction is 10, then the value of ΔG° will be $(R = 8JK^{-1}mol^{-1},\ T = 300\,K)$
Type	multiple choice
Option	$+5.527\,kJ\,mol^{-1}$
Option	$-5.527\,kJ\,mol^{-}$
Option	$+55.27\,kJ\,mol^{-1}$
Option	$-55.27\,kJ\,mol^{-1}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Reaction $H_2(s) + I_2(g) \rightarrow 2HI$; $\Delta H = 12.40$ kCal. According to this, heat of formation of HI will be-
Type	multiple choice
Option	12.40 kCal
Option	-12.40 kCal
Option	-6.20 kCal
Option	6.20 kCal
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The enthalpy of the formation of ammonia is -46.0 kJ mol^{-1}. The enthalpy change for the reaction $2NH_3(g) \rightarrow N_2(g)\ +3H_2(g)$ is:
Type	multiple choice
Option	$46.0\,kJ\,mol^{-1}$
Option	$92.0\,kJ\,mol^{-1}$
Option	-23.0 kJ mol^{-1}
Option	-92.0 kJ mol^{-1}
Answer	2
Solution	
Positive Marks	4
Negative Marks	1

Question	M is a metal that forms an oxide M_2O. $\frac{1}{2}M_2O \rightarrow M + \frac{1}{4}O_2 \quad \Delta H = 120\,kCal$. When a sample of metal M reacts with one mole of oxygen what will be the ΔH in that case
Type	multiple choice
Option	240 kCal
Option	-240 kCal
Option	480 kCal
Option	-480 kCal
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Heat of combustion of CH_4, C_2H_6, C_2H_4 and C_2H_2 gases are -212.8, -373.0, -337.0 and 310.5 kCal respectively at the same temperature. The best fuel among these gases is:
Type	multiple choice
Option	CH_4
Option	C_2H_6
Option	C_2H_4
Option	C_2H_2
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following equations corresponds to the enthalpy of combustion at 298 K:
Type	multiple choice
Option	$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$
Option	$2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(g)$
Option	$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
Option	$2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(l)$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1